S. No.	Concentration term	Expression	Illustration
4	Formality (F)	$\frac{\text{Number of Formula mass of solute}}{\text{Volume of solution (in L)}}$	5.85g of sodium chloride is dissolved in water and the solution was made up to 500 mL using a standard flask. The strength of the solution in formality is formality = $\frac{\text{Number of Formula mass of solute}}{\text{Volume of solution (in L)}}$ $= \frac{5.85}{58.5 \times 0.5\text{L}}$ $= 0.2$ F
5	Mole fraction (of a component) (x)	$\frac{\text{Number of moles of the component}}{\text{Total number of moles of all the components present in solution}}$ Consider a solution containing two components A and B whose mole fractions are x_A and x_B, respectively. Let the number of moles of the two components A and B be n_A and n_B, respectively. $x_A = \frac{n_A}{n_A + n_B}$ and $x_B = \frac{n_B}{n_A + n_B}$ Now, $x_A + x_B = \frac{n_A}{n_A + n_B} + \frac{n_B}{n_A + n_B} = 1$	0.5 mole of ethanol is mixed with 1.5 moles of water. The mole fraction of ethanol in the above solution is $= \frac{\text{Number of moles of the ethanol}}{\text{Total number of moles of ethanol and water}}$ $= \frac{0.5}{1.5+0.5} = \frac{0.5}{2.0} = 0.25$ The mole fraction of water in the above solution is $= \frac{\text{Number of moles of water}}{\text{Total number of moles of ethanol and water}}$ $= \frac{1.5}{2.0} = 0.75$ The mole fraction of water can also be calculated as follows mole fraction of water + mole fraction of ethanol = 1; mole fraction of water = 1 – mole fraction of ethanol = 1-0.25 = 0.75
6	Mass percentage (% w/w)	$\frac{\text{Mass of the solute (in g)}}{\text{Mass of solution (in g)}} \times 100$	Neomycin, aminoglycoside antibiotic cream contains 300 mg of neomycin sulphate the active ingredient, in 30g of ointment base. The mass percentage of neomycin is $\frac{\text{Mass of the neomycin sulphate (in g)}}{\text{Mass of solution (in g)}} \times 100$ $= \frac{0.3\text{ g}}{30\text{ g}} \times 100 = 1\%\text{ w/w}$

Unit 9 - jagan.indd 33 27-01-2021 15:06:22